

Retail Sales & Inventory Analysis

End-to-end retail sales and inventory analysis using Python, SQL, and Tableau to evaluate sales performance, profitability, supplier performance, and inventory risk.

NET SALES

\$10,482,110.25

GROSS PROFIT

\$4,634,359.15

GROSS MARGIN

44.21%

INVENTORY AT COST

\$1,096,577.97

Python

SQL

Tableau

KPI Storytelling

Business Context & Project Objective

Connecting retail sales performance with inventory risk to support better business decisions.

Business questions answered

- ◆ Which categories, products, stores, and suppliers drive sales and gross profit?
- ◆ Where is gross margin strongest or weakest across the product mix?
- ◆ Which products create slow-moving, stockout, or overstock risk?
- ◆ How can inventory decisions improve revenue protection and working capital?

Goal: turn raw retail sales and inventory files into reliable analytical outputs and clear business actions.

Project assets produced

1

Clean dataset

Final analytical dataset with standardized sales, profit, and inventory measures

2

Python notebook

Cleaning, transformation, feature creation, and export workflow

3

SQL analysis

Validation and business questions using structured queries

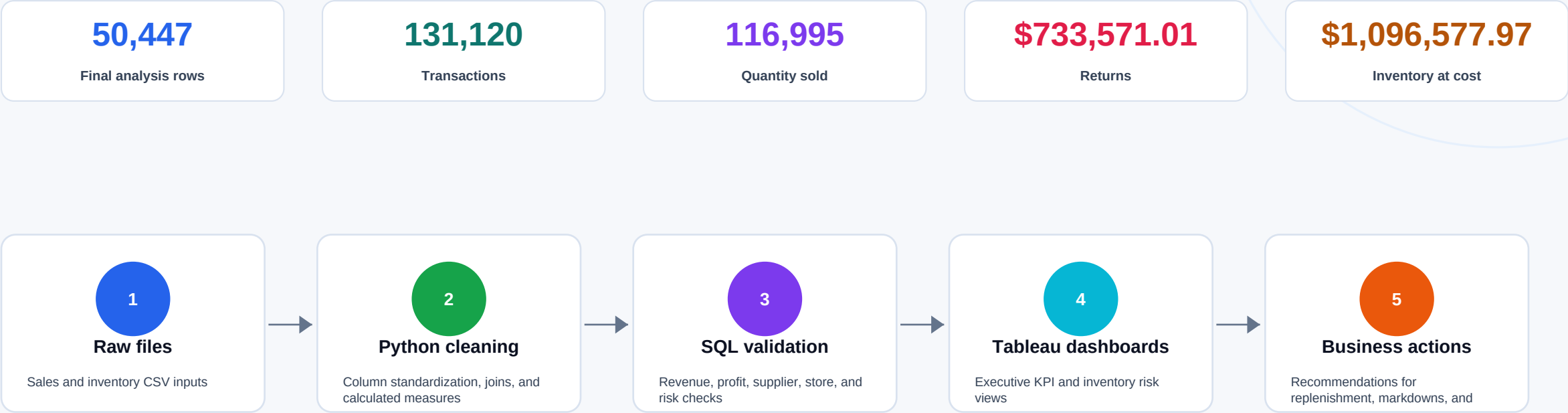
4

Tableau dashboards

Business Performance and Inventory Risk Analysis dashboards

Analysis Workflow

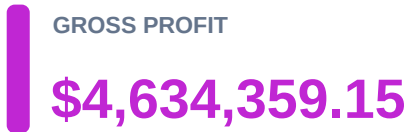
A reproducible pipeline from raw data to executive dashboards and recommendations.



Quality control focus: KPI consistency across Python output, SQL checks, and Tableau dashboards to keep results credible and reproducible.

Executive KPI Summary

High-level retail performance and inventory position.



Sales performance is strong

The business generated \$10.48M in net sales and \$4.63M in gross profit, supported by 131,120 transactions.



Margin is a key strength

A 44.21% gross margin indicates meaningful profitability, but low-margin product groups still need review.



Inventory requires active control

Current inventory at cost is \$1.10M, with slow-moving and overstock items creating working-capital pressure.

Response	Percentage
Doing a good job	68%
Not doing a good job	32%

KPI Overview																				
Current Inventory		Gross Margin%		Total Gross Profit		Total Net Sales		Total Qty Sold												
\$1,096,577.97		44.21%		\$4,634,359.15		\$10,482,110.25		116,995.00												
Net Sales by Product Category			Gross Profit by Product Category			Gross Margin % by Product Category														
Product Category			Product Category			Product Category														
Top 20 Products Sales			Top 20 Products by Gross Profit			Low Profitability Products														
Product Description			Product Description			Product Description														
Store Performance																				
		Store																		
Total Net Sa.		STR-1006	485.56K	399.70K	371.28K	346.39K	343.44K	335.34K	332.40K											
		STR-1229	399.70K	371.28K	346.39K	343.44K	335.34K	332.40K	329.19K											
		STR-1080	371.28K	346.39K	343.44K	335.34K	332.40K	329.19K	328.47K											
		STR-1194	346.39K	343.44K	335.34K	332.40K	329.19K	328.47K	327.53K											
		STR-1010	343.44K	335.34K	332.40K	329.19K	328.47K	327.53K	324.92K											
		STR-1235	335.34K	332.40K	329.19K	328.47K	327.53K	324.92K	303.17K											
		STR-1148	332.40K	329.19K	328.47K	327.53K	324.92K	303.17K	297.20K											
		STR-1094	329.19K	328.47K	327.53K	324.92K	303.17K	297.20K	284.13K											
		STR-1422	328.47K	327.53K	324.92K	303.17K	297.20K	284.13K	280.12K											
		STR-1044	327.53K	324.92K	303.17K	297.20K	284.13K	280.12K	274.41K											
		STR-1198	324.92K	303.17K	297.20K	284.13K	280.12K	274.41K	272.63K											
		STR-1253	303.17K	297.20K	284.13K	280.12K	274.41K	272.63K	270.73K											
		STR-1109	297.20K	284.13K	280.12K	274.41K	272.63K	270.73K	267.68K											
		STR-1203	284.13K	280.12K	274.41K	272.63K	270.73K	267.68K	258.64K											
		STR-1114	280.12K	274.41K	272.63K	270.73K	267.68K	258.64K	253.50K											
		STR-1107	274.41K	272.63K	270.73K	267.68K	258.64K	253.50K	253.08K											
		STR-1163	272.63K	270.73K	267.68K	258.64K	253.50K	253.08K	243.59K											
		STR-1157	270.73K	267.68K	258.64K	253.50K	253.08K	243.59K	241.74K											
		STR-1145	267.68K	258.64K	253.50K	253.08K	243.59K	241.74K	234.01K											
		STR-1038																		

Vendor 0166 and Vendor 0134 are the largest sales contributors, with \$3.29M and \$3.20M in net sales.

2

Leading product categories contribute the highest net sales and gross profit, making assortment planning important.

3

Total returns are \$733,571.01 and should be reviewed by product and supplier to identify avoidable return patterns.

Decision use: identify where to protect inventory, improve margin, and focus supplier conversations.

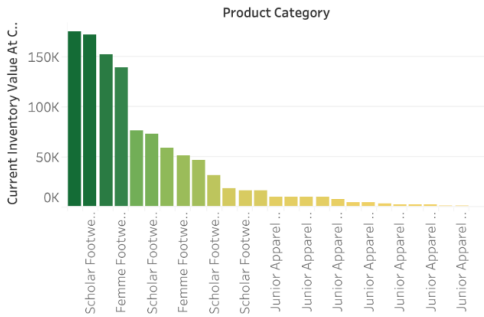
Inventory Risk Analysis Dashboard

Inventory risk indicators highlight stockout exposure, slow movement, and overstock pressure.

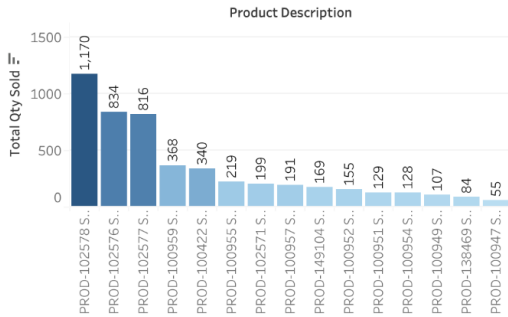
Inventory Risk Summary

Fast Moving Products	Slow Moving Products	Stockout Risk Products	Overstock Risk Products
13,997	21,658	10,323	21,585

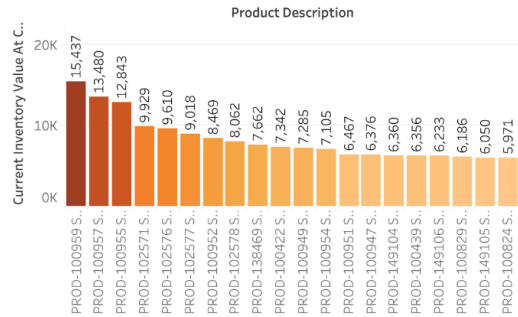
Inventory Value



Fast Moving Products



Slow Moving Products



Stockout Risk Products

PROD-102578 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-102577 SEG-500351 Scholar Footwear Flight Classic Zephyr				
PROD-100439 SEG-500064 Scholar Footwear Boot Collection Woodland	PROD-102574 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-102573 SEG-500351 Scholar Footwear			
PROD-102576 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-100438 SEG-500064 Scholar Footwear Boot Collection	PROD-100959 SEG-500123 Scholar Footwear	PROD-100422 SEG-500063 Scholar		
	PROD-102575 SEG-500351 Scholar Footwear Flight Classic	PROD-101707 SEG-500224 Scholar Footwear	PROD-100827 SEG-500112 Heritage		

Overstock Risk Products

PROD-100959 SEG-500123 Scholar Footwear Flight Classic Zephyr	PROD-102571 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-102578 SEG-500351 Scholar Footwear Flight Classic Zephyr			
PROD-100957 SEG-500123 Scholar Footwear Flight Classic Zephyr	PROD-102576 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-138469 SEG-5008729 Scholar Footwear Modern Legend Summit	PROD-149104 SEG-511422 Scholar Footwear		
PROD-100955 SEG-500123 Scholar Footwear Flight Classic Zephyr	PROD-102577 SEG-500351 Scholar Footwear Flight Classic Zephyr	PROD-100422 SEG-500063 Scholar Footwear Boot	PROD-100439 SEG-500064 Scholar Footwear		
	PROD-100952 SEG-500123 Scholar Footwear Flight Classic	PROD-100949 SEG-500123 Scholar Footwear Flight	PROD-149106 SEG-511422 Scholar Footwear	PROD-100824 SEG-500112 Heritage Footwear	

FAST MOVING PRODUCTS

13,997

SLOW MOVING PRODUCTS

21,658

STOCKOUT RISK PRODUCTS

10,323

OVERSTOCK RISK PRODUCTS

21,585

Decision use: prioritize replenishment for stockout-risk products and markdown/transfer actions for slow-moving or overstock items.

Recommended Business Actions

Targeted actions across revenue growth, margin protection, and inventory risk control.



Revenue focus

Protect inventory for the highest revenue categories and top stores to prevent lost sales in proven demand areas.



Margin protection

Investigate low-margin products and supplier/category combinations before scaling promotions or purchasing.



Inventory control

Reduce exposure in the 21,658 slow-moving and 21,585 overstock-risk products using markdown, transfer, or purchasing controls.



Replenishment priority

Monitor the 10,323 stockout-risk products more frequently and align restocking with sales velocity.



Supplier review

Use supplier rankings to validate dependency, negotiate performance terms, and improve product availability.



Dashboard cadence

Refresh KPIs regularly so sales, inventory, and supplier actions stay aligned with current performance.

This project demonstrates data cleaning, SQL analysis, Tableau dashboarding, KPI validation, and business insight communication.